Fiscal Year:	FY 2005	**Task Last Updated:**	FY 04/21/2006
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Project Title:	Long-Term Simulation Microgravity Inhibits Spermatogenesis in Adult Male Rats		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology (2) Developmental Biology		
Space Biology Special Category:	None		
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Zip Code:	66160-0001	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2000 Space Biology 00-OBPR-01
Start Date:	07/01/2001	**End Date:**	06/30/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Tash, Joseph		
COI Name (Institution):			
Grant/Contract No.:	NAG 2-1491		
Performance Goal No.:			
Performance Goal Text:			
	NASA's goals in developmental biology include how gravity may affect the capacity of organisms to reproduce. The overall hypothesis for this project is that extended microgravity (µG) exposure inhibits spermatogenesis in adult rats. Earlier studies from our laboratory examined the effect of µG on protein phosphorylation coupled to sperm motility in µG. In two Space Shuttle experiments we found that I) flagellar protein phosphorylation during activation of sea urchin sperm motility occurs faster in µG than in 1G, and ii) temporal changes in protein phosphorylation and dephosphorylation in response to egg chemotactic factors are altered in µG compared to 1G. Follow-up studies showed that sperm-egg binding and egg fertilization were significantly inhibited at 2G. These results suggest that sperm functions including fertilization are sensitive to changes in the gravitational environment. In order to focus on vertebrates, we recently examined whether spermatogenesis is impacted in long-term simulated µG using the hindlimb suspended (HLS) rat model. HLS is widely utilized to reproduce and examine changes in bone and muscle associated		

Task Description:	with µG. No spermatogenic cells beyond round spermatids were present in scrotal testes in 6-week HLS adult male rats even though no changes in FSH, LH, or corticosterone and only slight changes in circulating testosterone were observed. The following specific aims will be pursued to address the hypothesis: 1. Determine and characterize the time-dependent decline in spermatogenesis in HLS adult male rats 2. Determine whether the decline in spermatogenesis in HLS males is reversible 3. Elucidate the physiologic mechanism(s) underlying the decline in spermatogenesis in the HLS model Two of the Bioastronautics Critical Roadmap Risks are highlighted by our research: • Risk 28: Increased cancer morbidity or mortality risk in astronauts may be caused by occupational radiation exposure – May be influenced by other space flight factors including microgravity & environmental contaminants. Mars mission not feasible unless improved shielding is developed. – Exposure to space radiation increases the risk of developing cancer later in life. • Risk 30: Chronic and Degenerative Tissue Risks – Radiation exposure may result in degenerative tissue diseases. May be caused by occupational radiation exposure or the combined effects of radiation & other space flight factors – Acute or chronic illness due to tissue degeneration may lead to mission impacts, or adverse health consequences after return. A key question raised by our results is whether the long term hindlimb unloaded model is predictive of the effect of long term microgravity on male reproductive health. This health concern is a separate but related issue to the increased risk of gonadal cancer due to radiation exposure during space flight. It should be noted that there is extensive literature demonstrating that the risk of gonadal cancer is increased by hyperthermia in the cryptorchid model24, 25. It is difficult to ascertain whether the time interval between testicular hyperthermia and the occurrence of testicular cancer would be within the time frame of a 30 month mission to Mars. In this regard, the onset of testicular cancers is postulated to be triggered in the post-cryptorchid male after puberty because of the influence of the hormonal milieu. Thus in the adult male with chronic testicular hyperthermia, this interval could be decreased because the adult male hormonal milieu is already present. When combined with increased radiation exposure, this will significantly elevate the risk for testicular cancer (at least 4-10 fold based on the cryptorchid model). In any case, because of the lack of data or investigations focused on the conditions experienced in space flight, these questions warrant more detailed investigation under controlled conditions. This concern raised by our studies is relevant for human male astronauts in the context of the Bioastronautics Critical Roadmap as well as for rodents as an experimental model for humans. Unfortunately, no animal data is available because the 6 weeks hindlimb unloading is yet to be paralleled by space flight experiments of at least equivalent length. Human data from male astronauts or cosmonauts exposed to Microgravity for longer than Space Shuttle mission duration has not been published or been made available in response to gather such information. Thus it is impossible to ascertain whether or not similar problems have been reported in humans exposed to long term microgravity conditions. A key question raised by these studies is whether the hindlimb unloaded model is relevant as a model for microgravity in males with scrotal testes. The regulation of testicular temperature well below body temperature is critical for the maintenance of spermatogenesis. The testicular temperature is regulated by a combination of blood flow and position relative to the body. Two opposing forces regulate the position of the testis: 1) the gravitational vector pulls the testicular mass away from the body promoting testicular cooling; 2) the cremasteric muscles pull the testes towards the body thereby increasing testicular temperature. This mode of temperature regulation is identical in all scrotal males including but not limited to humans, primates and rodents. Absence of the gravitational pull on the testis in microgravity eliminates the opposing affect of the cremasteric muscles causing the testes to remain elevated near the body; this is mimicked in the hindlimb unloaded animal because the head downward tilt position prevents gravity from pulling the testis away from the body. Thus microgravity and hindlimb unloading both prevent the testes from cooling properly, causing testicular hyperthermia and compromised spermatogenesis. Thus, we hypothesize that in long term space flight, chronic testicular hyperthermia may occur in both rats and humans. In relation to the BCPR, this can cause significantly increased health risks not only for male infertility or sterility, but also for increased testicular cancer due to the well known combined effect of testicular hyperthermia and radiation exposure. Ultimately, this model must be verified/tested by long term rodent and human space flight experiments, developing appropriate countermeasures, and monitoring male astronaut health post flight. We are thus faced with the following biomedical question: what level of post-flight astronaut cancer and/or infertility risk is acceptable? These risks will magnify with the lengthening duration of space flight and exposure to radiation outside of low earth orbit. The problem of gonadal cancer could even occur within the time frame of a long duration mission such as the 2 year Mars expedition, and post-flight following missions of 6 months or longer on ISS or more likely to the moon. Sterility and subfertility would be manifested post flight in such cases. Is there a willingness of NASA to address these questions by animal and human experimentation, and to address countermeasure development and assessment?
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	NASA's goals in developmental biology include how gravity may affect the capacity of organisms to reproduce. The overall hypothesis for this project is that extended microgravity (µG) exposure inhibits spermatogenesis in adult rats. Earlier studies from our laboratory examined the effect of µG on protein phos-phorylation coupled to sperm motility in µG. In two Space Shuttle experiments we found that i) flagellar pro-tein phosphorylation during activation of sea urchin sperm motility occurs faster in µG than in 1G, and ii) temporal changes in protein phosphorylation and dephosphorylation in response to egg chemotactic factors are altered in µG compared to 1G. Follow-up studies showed that sperm-egg binding and egg fertilization were significantly inhibited at 2G. These results suggest that sperm functions including fertilization are sensi-tive to changes in the gravitational environment. In order to focus on vertebrates, we recently examined whether spermatogenesis is impacted in long-term simulated µG using the hindlimb suspended (HLS) rat model. HLS is widely utilized to reproduce and examine changes in bone and muscle associated with µG. No spermatogenic cells beyond round spermatids were present in scrotal testes in 6-wk HLS adult male rats even though no changes in FSH, LH, or corticosterone and only slight changes in circulating testosterone were ob-served. The following specific aims will be pursued to address the hypothesis: 1. Determine and characterize the time-dependent decline in spermatogenesis in HLS adult male rats 2. Determine whether the decline in spermatogenesis in HLS males is reversible 3. Elucidate the physiologic mechanism(s) underlying the decline in spermatogenesis in the HLS model These studies will provide important new data concerning the impact of extended exposure to µG on the ability of males to reproduce as well as critical environmental impact data for astronauts working for ex-tended periods

	on the International Space Station and the planned Mission to Mars.
Task Progress:	Whether species can survive multiple generations in microgravity (μG) was one question driving NASA biological research at the inception of this project. Hindlimb unloading by suspension (HLS) is used to model the effects of μG on mammals. We showed aspermatogenic sterility in rats 6 months after terminating a 6 week HLS treatment. All HLS males exhibited mating behavior but were sperm negative and produced no litters. Control males that also were tail-harnessed and partially ligated to prevent abdominal testes like HLS, but not hindlimb-elevated, maintained spermatogenesis and produced normal litters. HLS testes were half the mass of controls and either Sertoli cell only, or fibrotic and lost both Sertoli cells and germ cells, and showed massive seminiferous apoptosis. Analysis of testes during the initial 6 week HLS treatment revealed that infiltration of myeloperoxidase-containing inflammatory cells precedes the apoptitic loss of spermatogenesis after 2-3 weeks of HLS. These results raise concerns that long-term spaceflight may compromise male fertility.
Bibliography Type:	Description: (Last Updated: 01/24/2022)
Abstracts for Journals and Proceedings	Tash, J.S.; Wolfe, S.; Bilgen, M. "Magnetic Resonance Imaging (MRI) Study of Anatomical Features Underlying Sterility in Hindlimb Unloaded (HLS) Male Rats." American Society for Gravitational and Space Biology annual meeting, November 2004, New York, NY. Grav Space Biol Bull. 2004 Nov;18(1):57. , Nov-2004